



COLLEGE
OF
MEDICINE

TREATMENT OF INFERTILITY & IVF

:Presented by

Dr. ROZHAN YASSIN KHALIL

Assistant professor & consultant obsgyn

FICOG, CABOG, HDOG, FICS, MBChB

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:Objective

- • Understand the provision and regulation of fertility treatment.
- • Explain the processes and procedures involved in assisted reproductive treatment (ART).
- How you deal with male factors infertility.
- Does life style change affect infertility treatment.
- Infertility treatment need follow up and good privacy.

Treatment of infertility

● 1. General advice:

- 1. All couples trying for a pregnancy will benefit from some general advice such as **cessation of smoking and limiting alcohol intake.**
- 2. Pre-treatment counselling should include advice about **general lifestyle measures including the need to achieve an optimum BMI.**

:General advice

- 3. This will involve weight loss in women with a BMI of over 30, but may require **some women with weight-related amenorrhoea and anovulation to gain weight.**
- 4. Periconceptual dietary supplementation of **folate** has been shown to reduce the risk of neural tube defects.

TREATMENT OF.2

:ANOVULATION

Women should be made aware of potential risks of multiple pregnancy and ovarian hyperstimulation, when prescribing ovulation induction drugs.

Male factor problems and tubal pathology should be excluded.

In the absence of a history suggestive of tubal disease it may be reasonable to defer formal tubal patency tests for three cycles to allow less invasive treatment options such as clomifene to be used.

TREATMENT OF ANOVULATION

- In women with weight loss associated amenorrhoea, **treatment should be deferred until a target BMI of 20 kg/m² is reached.**
- The most physiological treatment of **anovulation or hypothalamic amenorrhoea** is with **pulsatile administration of GnRH agonists administered**

TREATMENT OF :ANOVULATION

- **Weight loss should be the first line of treatment in obese women with anovulation due to PCOS.**
- Central obesity and high BMI are important predisposing factors for insulin resistance, hyperinsulinaemia and hyperandrogenaemia

TREATMENT OF ANOVULATION

- Effective treatment of obesity can reverse these effects and facilitate the effects of ovulation induction agents.
- In obese women with PCOS a loss of **5–10%** of body weight may be enough to restore reproductive function in **55–100%** women within **6 months**.

:Clomifene citrate.1

- **Clomifene citrate (clomide):** is an orally active synthetic non-steroidal compound with **oestrogenic as well as anti-oestrogenic properties**, which has traditionally been the treatment of choice in women with anovulatory PCOS.
- It displaces oestrogen from its receptors in the hypothalamic-pituitary axis, **reduces the negative feedback effect of oestrogen and encourages GnRH secretion.**

:Clomifene citrate

- It is administered in an initial daily dose of **50 mg on days 2–6** of a spontaneous or induced menstrual period.
- The dose can be increased by **50 mg** per day till ovulation is achieved, up to a maximum of **150 mg per day**.
- A course of **6** cycles can be used in women who respond to the drug.

:Clomifene citrate

- It is necessary to monitor follicular response, with TV scans to minimize the risk of multiple pregnancy.
- Ovulation is expected to occur in **80%** and pregnancy in **35–40%** women on clomifene.
- Approximately 20–25% of women show no response to clomifene citrate and are considered to be resistant.**

:Adverse reactions of clomifene

- 1. Anti-oestrogenic effects include thickening of cervical mucus and **hot flushes in 10% of women.**
- 2. Other side effects: abdominal distension (2%), abdominal pain, nausea, vomiting, headache, breast tenderness and **reversible hair loss.**
- 3. Clomifene has a **mydriatic action** that can result in **blurred vision and scotomas in 1.5%** of women. These changes are reversible.

Adverse reactions

- 4. Significant ovarian enlargement occurs in **5%** but ovarian hyperstimulation syndrome (OHSS) is rare(**<1%**).
- 5.The multiple pregnancy rate associated with clomifene is **7–10%**.
- 6. link with ovarian cancer has been described in women receiving more than **12(better 6)** cycles of clomifene.



Tamoxifen.2

- Tamoxifen has a structure similar to clomifene.
- The recommended dose is **20–40 mg** per day from **day 2, for 5 days**.
- Pregnancy rates with tamoxifen are similar to those obtained with clomifene and it may have a less potent anti-oestrogenic action on cervical mucus.
- Another treatment (**letrozole**) aromatase inhibitor.

Gonadotrophins.3 :

- Treatment with gonadotrophins is contemplated when women either do not respond to clomifene or fail to conceive after **6–12 ovulatory cycles**.
- Preparations in common use include **recombinant FSH or purified urinary human menopausal gonadotrophin which contains FSH as well as LH**.

Metformin.4

- The strong association between anovulation and insulin resistance/hyperinsulinaemia has led to speculation that lowering insulin levels would lead to improvement in the clinical and metabolic profile of women with **PCOS**.
- While this could be achieved by weight loss alone, an insulin sensitizing agent like **metformin** was felt to be particularly suitable.
- **Metformin is an oral biguanide**

:Laparoscopic ovarian drilling

- Laparoscopic ovarian drilling (LOD) by diathermy or laser is a further treatment option for women with anovulation associated with PCOS.
- The procedure appears to be more successful in women who are **slim and have high LH levels**; the mechanism for its effect is unknown.
- A unipolar coagulating current is used to deliver four punctures to a depth of 4mm in each ovary.
- **The principal advantages of ovarian drilling include monofollicular ovulation resulting in fewer multiple pregnancy rates.**

HYPERPROLACTINAEMIA

- Prolactin secretion is regulated by the tonic inhibitory control of dopamine.
- **Bromocriptine**: which has a dopamine like action, is effective in hyperprolactinaemia.

It shrinks **80% of macroadenomas**, and can help to normalize prolactin values in **80–90%** and restore ovulation in **70–80%** of women

bromocriptine

- Long-term treatment with bromocriptine results in **pregnancy rates of 35–70% per woman**. Due to its short half-life, bromocriptine needs to be **administered two to three times a day**.
- Side effects including **nausea, headache, vertigo, postural hypotension, fatigue and drowsiness** can be minimized by initiating treatment with a low dose of bromocriptine (1.25 mg) at bed time with a snack.

Cabergoline and Quinogolide

- Cabergoline and *Quinogolide* are newer dopamine agonists which recently licensed for treatment of hyperprolactinaemia.
- Fewer side effects and longer half-lives allow a once daily dose for **quinogolide** and a twice weekly dose for Cabergoline.

Management of male factor :infertility

- **1. General measures should include advice about stopping smoking and reducing alcohol consumption.**
- Where a specific cause is identified, targeted treatment should be considered.
- **2. In the majority of cases no cause for abnormal semen parameters can be identified, and assisted reproduction offers the only option for men to have their own genetic offspring: include:**



INTRAUTERINE.1

:INSEMINATION

- **Intrauterine insemination (IUI):**
using washed sperm may be considered in cases where semen parameters show mild or moderate abnormalities

IVF/ICSI.2 :

- Where semen parameters are poor, it may be appropriate to consider IVF treatment straightaway.
- In men with grossly reduced sperm concentrations (**below 5 million/ml**) ICSI is the treatment of choice.
- Obstructive azoospermia, in the presence of normal testicular volume and FSH levels can be treated by surgical sperm retrieval followed by ICSI.
- The prognosis for **non-obstructive azoospermia** associated with small atrophied testes and high FSH levels in poor and donor insemination (DI) may need .

3.DONOR INSEMINATION:

- Where surgical sperm retrieval is not possible, or when ICSI is not feasible, **insemination of thawed frozen donor sperm** may be considered.
- Donors are screened for hereditary conditions and blood-borne viruses.

:Other interventions in male factor infertility.4

1. Gonadotrophins have no role in enhancing pregnancy rates in men with idiopathic oligozoospermia.
2. Other interventions which have been shown to be ineffective include anti-oestrogens (**clomifene and tamoxifen**), androgens, bromocriptine.
3. **Antioxidants (Vitamins C and E and glutathione)** can improve semen parameters in men.

SURGICAL TREATMENT .4

- Data on success rates after surgical procedures for post infective block, including epididymovasotomy, are limited.
- Observational studies have described a **post-surgical patency rate of 52% and a pregnancy rate of 38%.**

Treatment of tubal factor infertility

- The majority of women with **moderate to severe tubal damage are unlikely to conceive spontaneously** and IVF is generally accepted as the treatment of choice in these cases.
- techniques for surgical reconstruction of damaged or occluded tubes have described.
- Higher pregnancy rates reported in women who underwent tubal surgery than in those who did not.

Types of assisted conception

- There are many types of assisted conception available in the modern unit.
- These range from **less invasive procedures** such as **IUI** through to the widely known **IVF**, with or without **ICSI**.

Intrauterine insemination

- Intrauterine insemination (IUI) is where a prepared sample of sperm is inseminated into uterine cavity at the appropriate time of the patient's menstrual cycle.

Approximately **two weeks** later a pregnancy test is performed to see if the cycle has been successful

:IUI

- Success rates increase from **unstimulated IUI through to stimulation with Clomid and FSH.**
- The overall success rate, as with any subfertile couple, depend on multiple factors , most importantly female age and with IUI the quality of the sperm.

Though IUI can be used for mild male factor problems, Success rates for stimulation with Clomid and then generally accepted levels of **between 12 and 18%** per cycle when FSH is used in the protocol.

PROTOCOLS:

- IUI can be performed in a natural cycle, with Clomid alone, with **Clomid** and then **FSH** injection or purely with FSH.
- a single human chorionic gonadotrophin (hCG) injection approximately **36 hours** prior to the insemination to ensure optimal timing with ovulation.

:COMPLICATIONS of IUI

- The main complication of IUI occurs when FSH has been used and this is higher order multiple births.
- Most centres would expect a twinning rate of between **10 and 15%** .
- this is normally due to inadequate monitoring and inadequate numbers of cycles being cancelled .



ADVANTAGES

- IUI is relatively a simple technique that is cost-effective and can be offered by both secondary and tertiary fertility centres.
- **It is not as invasive as IVF and allows fertilization to occur within the fallopian tubes and therefore is generally acceptable to most religious groups.**



:DISADVANTAGES

- The success rates are lower than those with IVF.
- if the cycle fails less information is obtained than with an IVF cycle – particularly pertaining to possible egg or subsequent embryo quality.
- It also requires at least one healthy fallopian tube and reasonable sperm parameters.
- significant increase in higher order multiple birth with expected sequelae of these.

:INDICATIONS of IUI

- 1. • ● Unexplained infertility
- 2. • ● Mild male Factor
- 3. • ● Ejaculatory problems
- 4. • ● Cervical problems
- 5. • ● Ovulatory disorders
- 6. • ● Mild endometriosis
- 7. • ● optimize the use of donor sperm .

.I.V.F

In

Vitro

Fertilization

“Test tube babies”



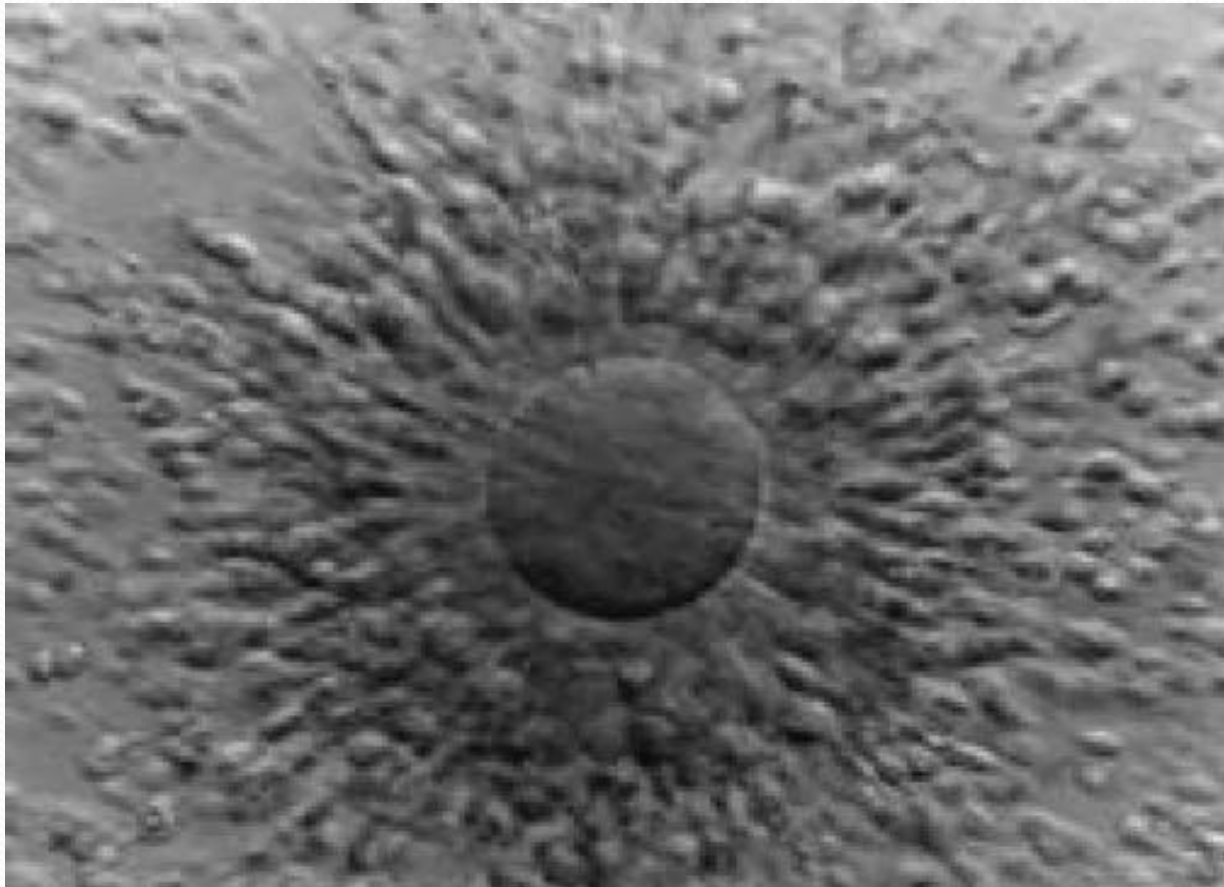
Louise Brown, 1978



In vitro fertilization

- IVF is where the mature oocyte is surgically removed from the ovary and then fertilized with sperm in the laboratory.
- The world's first successful IVF baby was delivered by Patrick Steptoe in **1978**.

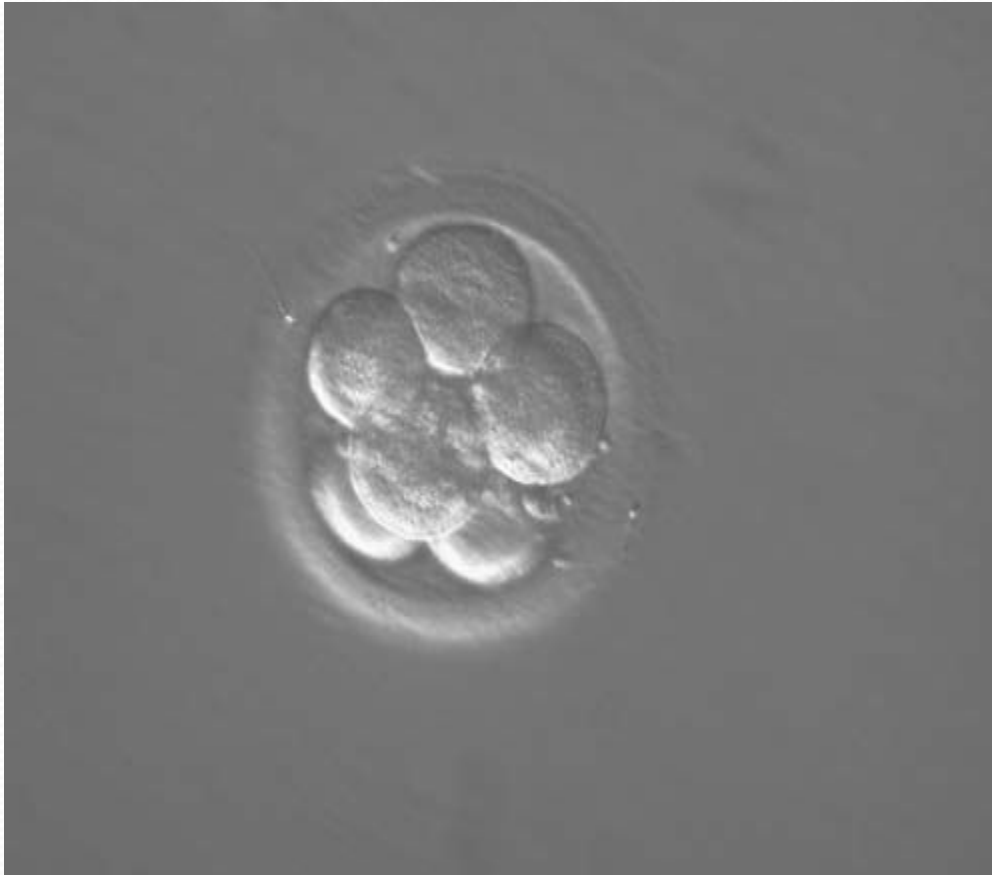
:hUman oocyte



Human embryo :2 pronuclus



Day 4 morula





Over the last 25 years the success rates and types of IVF have greatly improved and at present there are well over **4 million babies born throughout the world by this technique**

:INDICATIONS of IVF

- •1. Severe tubal disease – tubal blockages
- • 2. Severe endometriosis
- •3. Moderate male factor
- • 4. Unexplained infertility
- •5. Unsuccessful IUI

:PROTOCOLS of IVF

- 1. Initially simple forms of ovulation induction using Clomid and human menopausal gonadotrophins (**hMG's**) were used.
- protocols broken down into three main categories:
 - 1 Natural Cycle
 - 2 Long protocol – Agonist cycles
 - 3 Short protocol – Antagonist cycles

:MONITORING.2

- It is essential that adequate monitoring is performed during stimulation of the ovaries with exogenous gonadotrophins.
- Serial transvaginal ultrasounds to assess the follicular growth should be used



INJECTION.3

- This is used to induce final maturation of the oocytes prior to the oocyte retrieval.
- **10,000** units of urinary hCG is generally used although in patients with an over response this can be decreased down to 5000 units.
- hCG should be given when either one or two lead follicles have reached **18mm**.
- The injection is normally given around midnight to allow for oocyte retrieval approximately **34 hours** later prior to physiological ovulation occurring.

:OOCYTE RETRIEVAL.4

- Originally, this was done laparoscopically but with the advent of real-time ultrasound this allowed a less invasive oocyte retrieval by ultrasound directed needling of the ovaries.
- particularly with the advent of transvaginal (TV) scanning, has allowed both the monitoring of the ovary during stimulation and the actual retrieval itself to be Done transvaginally .

EMBRYO TRANSFER.5

- Eggs are fertilized either by routine insemination with a concentration of approximately 100,000 normally motile sperm per ml or by ICSI.
- They are incubated in a commercially prepared culture medium under strict laboratory conditions. Not only is the **temperature carefully controlled within the incubators but also the gas content and pH.**

- Most embryos are transferred at **day 2** post egg collection.
- More embryos are now being transferred on **day 5**, at the **blastocyst stage**.
- Approximately **55 to 60%** of all mature eggs fertilize normally and then these are graded
- Embryo transfer should be performed under ultrasound guidance as this allows more accurate placement of the embryos in the uterine cavity .

- Although there is no chance that the embryos can ‘fall out’,
- many patients are not surprisingly very cautious at this stage and quite often are **allowed to rest in a supine position for up to 2 hours** before being allowed to leave the hospital.



LUTEAL PHASE.6

SUPPORT

- luteal phase support (LPS) is generally thought necessary.
- Although natural cycle IVF does not need this, supraovulation may impair normal corpus luteal function and the use of LPS has been shown to improve success rates .
- but pregnancy rates without it are generally thought to be significantly lower .
- **LPS is broadly divided into two types:**

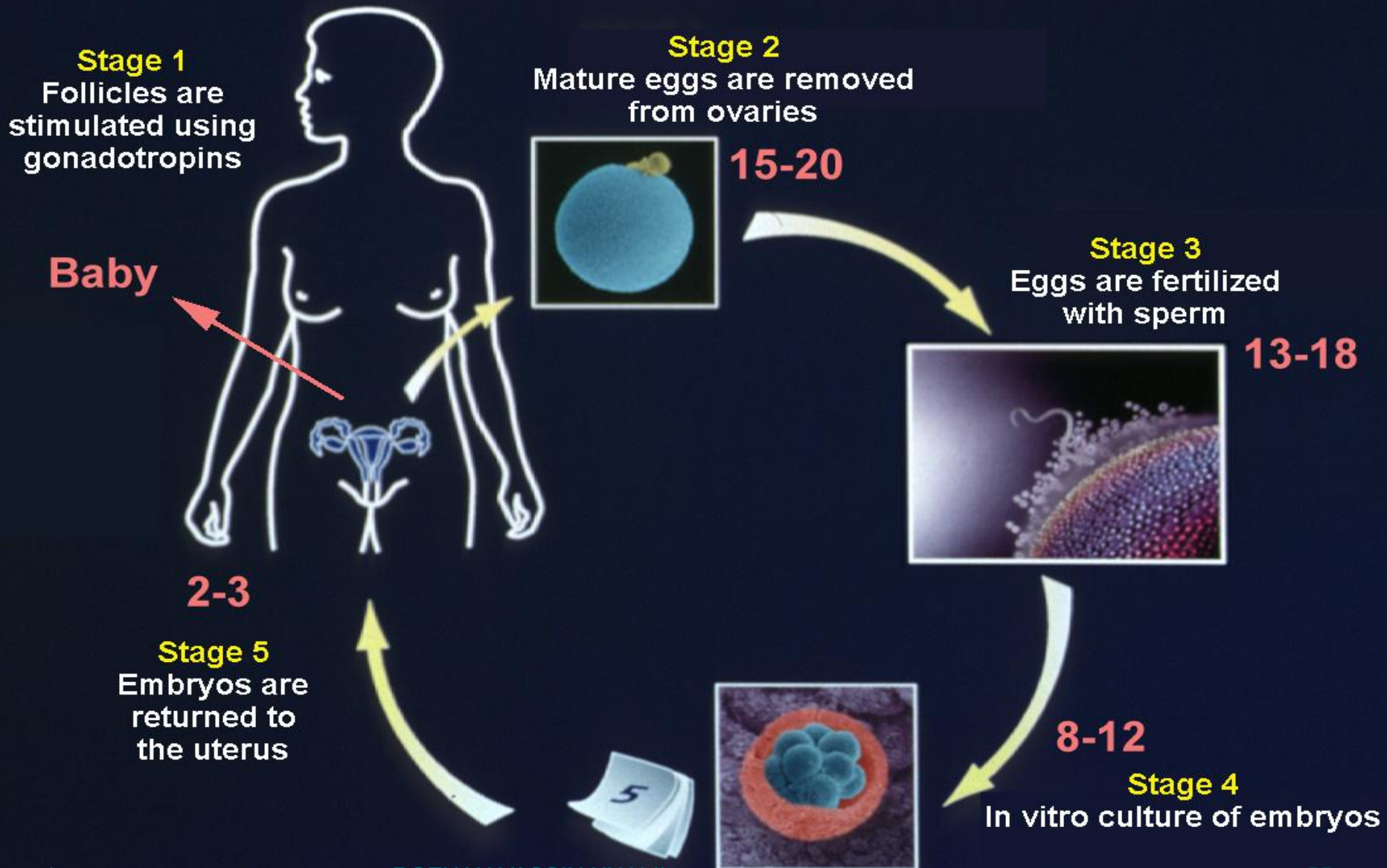
- **1.first the use of luteolytic preparations, such as hCG**
- **2. second, the use of progestogens or progesterone.**
- hCG is given by a subcuticular injection in small aliquots that stimulates the patient's own ovaries to produce more progesterone.
- It has been shown to be equally efficacious as progesterone but does require an injection

- The use of progesterones is more common and these can be given as **tablets, injections or vaginal pessaries/rectalsuppositories**.
- Intravaginal or rectal use of progesterone achieves extremely good tissue levels very rapidly.
- It is known that LPS should be given for a minimum of 2 weeks, but some clinics routinely use up to **12 weeks**.

PREGNANCY TEST.7

- ◎ The wait between the embryos being replaced and the pregnancy test is the most psychologically stressful time for the majority of patients.
- ◎ Generally pregnancy tests are performed around **12 days** from the embryo transfer with a serum pregnancy test.
- ◎ then it is usual to offer the patient a transvaginal scan **2 to 3 weeks** later to ensure that the pregnancy is intrauterine and also to assess its viability.

IVF- Clinical Processes



Complications of assisted conception

- **1. Multiple births**
- The most common complication of assisted conception is that of multiple births,
- approximately **25%** have twins when 2 or 3 embryos are transferred.
- In the UK the maximum of 3 embryos can be transferred

- **2.Ovarian hyperstimulation syndrome**
- Ovarian hyperstimulation syndrome (OHHS) can occur in any IVF cycle, but usually is only mild to moderate.
- Severe OHHS can be life threatening and should happen in less than 2% of cases.
- It generally occurs in specific at-risk groups, in particular in young patients who have polycystic ovaries.

- **3.Ectopic pregnancies:**

Ectopic pregnancies can occur with any of the assisted reproductive techniques.

- In IVF programmes the generally accepted rate is between **2 and 5%**.

Ovarian hyperstimulation syndrome (OHSS)

- **Ovarian hyperstimulation syndrome (OHSS) :**
can occur when drugs are used to stimulate the ovaries to produce the eggs needed for fertilization in A.R.T (IVF) treatment

Ovarian hyperstimulation syndrome (OHSS)

Three grades of OHSS can be diagnosed clinically :and sonographically

Mild OHSS : is associated with mild abdominal -1 discomfort, and the ovaries are usually less than **5 cm** .in diameter on sonographic examination

moderate OHSS: the ovaries measure between **5 -2 and 10 cm** & associated with more abdominal pain

severe OHSS: the ovaries are greater than **10 cm** in -3 diameter & characterized by the presence of free intraperitoneal fluid (ascites) due to increased vascular .permeability

Pleural effusions, hypotension, and oliguria have also been .described

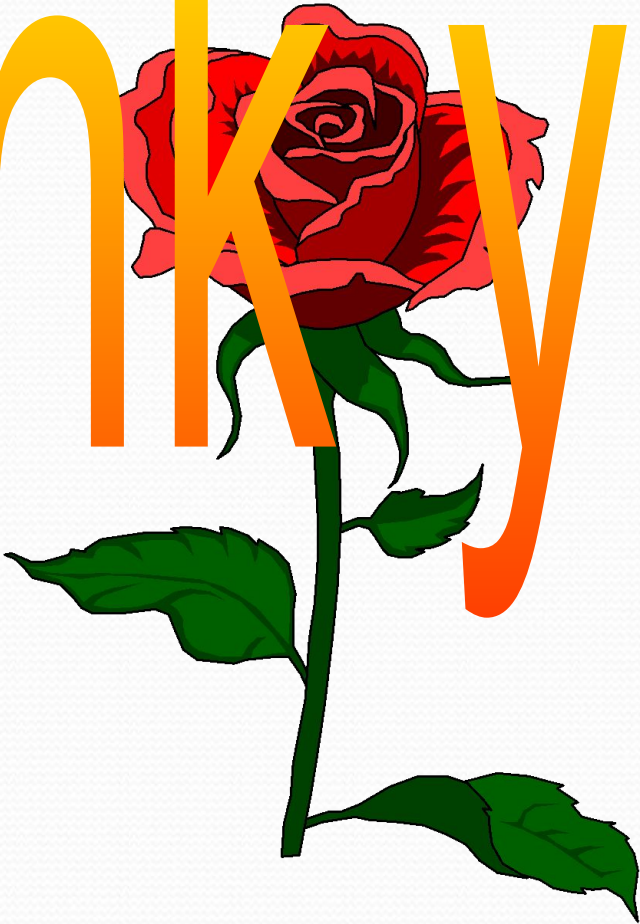
The patient also may have fluid and electrolyte disturbances and is at increased risk for torsion of the enlarged ovaries& .increased risk for **thrombotic** problems

Severe OHSS occurs in fewer than 2% of patients. this can be .life-threatening

:Conclusions

- The key to effective management of infertility is an appreciation of the likelihood of spontaneous pregnancy in a couple.
- The availability of assisted reproduction techniques has revolutionized the outlook for many couples with refractory infertility.
- The challenge now is to improve treatment success while minimizing risks.

Thank you

A detailed illustration of a red rose with green leaves and a stem, positioned behind the word 'Thank' in the 'Thank you' text.